



Hands-On Activity: Introduction to BigQuery



Activity overview

You have recently been introduced to BigQuery, a data warehouse on Google Cloud that data analysts use to query, filter large datasets, aggregate results, and perform complex operations. In this activity, you will explore the BigQuery interface; upload public data to your console; and write some simple SQL queries using **SELECT**, **FROM**, and **WHERE**.

By the time you complete this activity, you will be more familiar with writing queries in the BigQuery interface. This will enable you to practice SQL, which is important for working with databases in your career as a data analyst.



Step-By-Step Instructions

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

Step 1: Get a BigQuery account

For this activity, you will need a BigQuery account. If you haven't made one already, follow the instructions from the [Using BigQuery](#) reading.

Once you have your account, start exploring!

Step 2: Open your BigQuery console

1. Log in to [BigQuery](#).
2. Select the **Go to console** button on the BigQuery homepage. This will open a new tab with your console.

BigQuery

Benefits

Key features

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Migrating data warehouses to BigQuery

Predictive analytics

Bring any data into BigQuery

All features

BigQuery

Serverless, highly scalable, and cost-effective multicloud data warehouse designed for business agility.

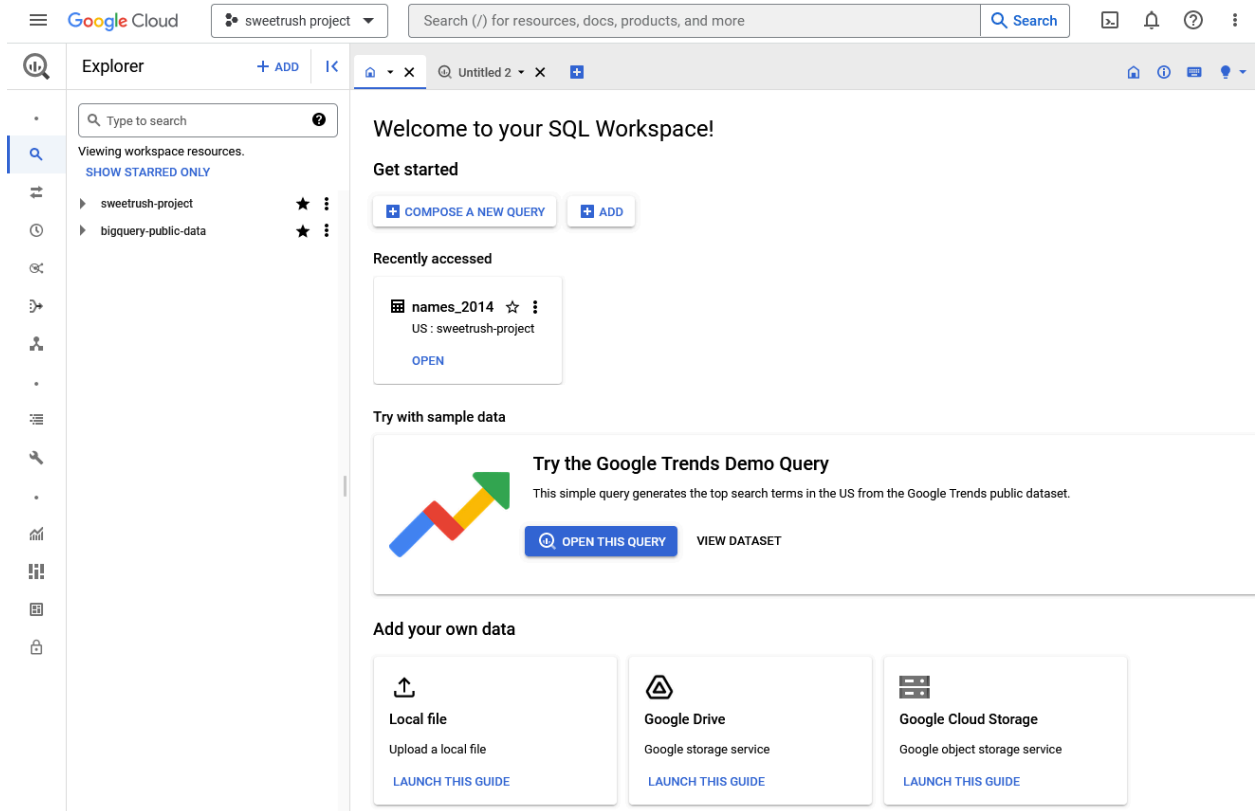
New customers get \$300 in free credits to spend on Google Cloud during the first 90 days. All customers get 10 GB storage and up to 1 TB queries/month, completely free of charge.

[Go to console](#)

[Contact sales](#)

- ✓ Democratize insights with a secure and scalable platform with built-in machine learning
- ✓ Power business decisions from data across clouds with a flexible, multicloud analytics solution
- ✓ Run analytics at scale with 26%–34% lower three-year TCO than [cloud](#)

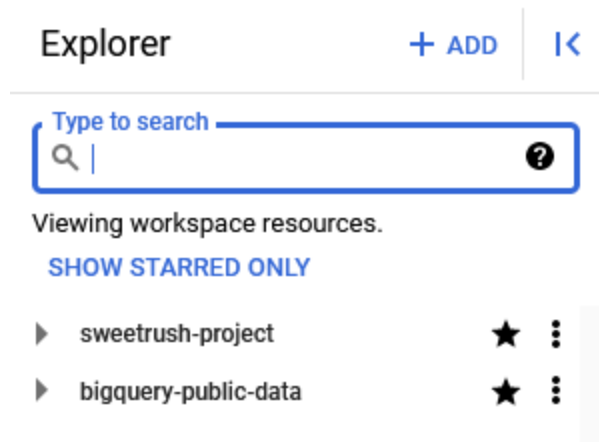
3. Take a moment to explore your console. The **Explorer** menu includes a search bar you can use to find resources, pinned projects, and the **+ ADD** button for adding data. The Editor welcome page is where you will navigate to a query editor, try sample data, add local data files, add Google cloud storage, or add other external connections. You can also find your job history, query history, and saved queries here.



Step 3: Access public data in BigQuery

In order to start writing queries, you will need some data to work with. Once you're familiar with the BigQuery interface, you can access a public dataset directly from your console.

1. Select the search bar in the **Explorer** pane.



2. Enter “london bicycle” in the search box and press enter; this will return the `london_bicycles` database from the Greater London Authority. Select the database for more details. If you cannot find it, make sure you're searching in all projects. The `london_bicycles` database is in the `bigquery-public-data` project.

Explorer + ADD <

Type to search × ?
Q london bicycle

Found 2 results.
[SEARCH ALL PROJECTS](#)

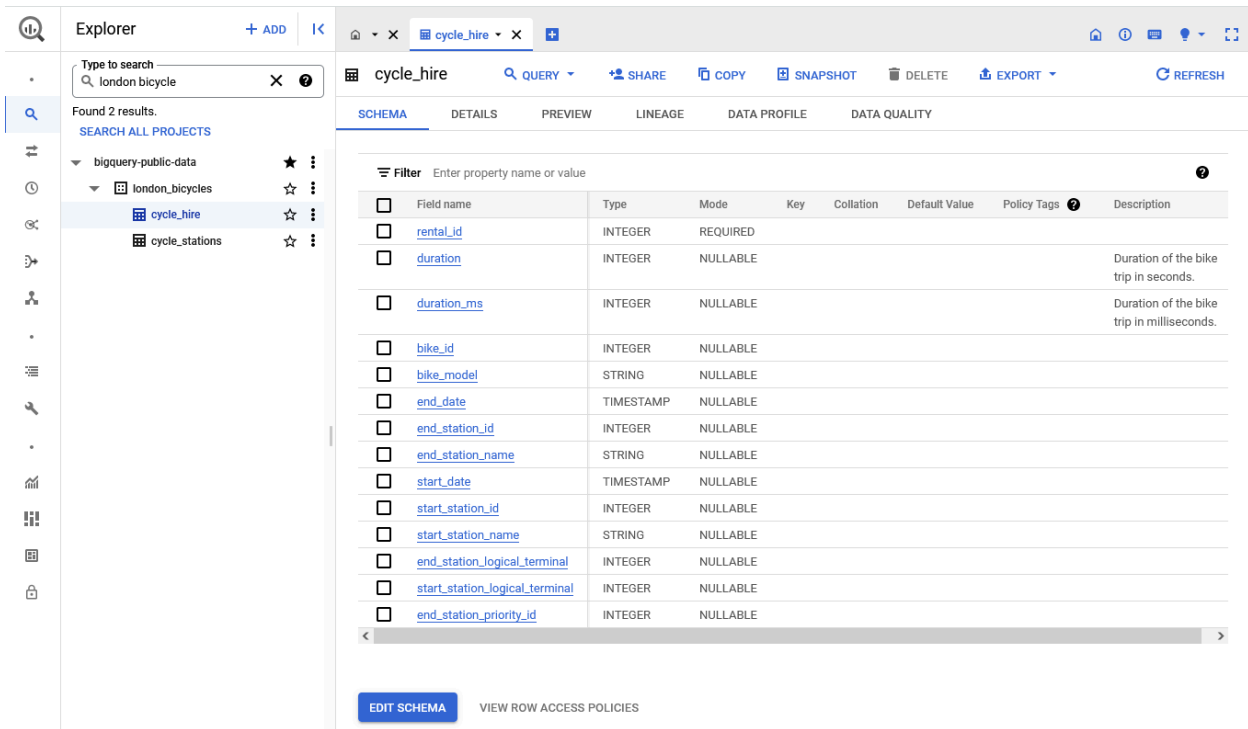
▼ bigquery-public-data ★ ⋮
▶ 🗃️ london_bicycles ☆ ⋮

🗃️ london_bicycles +

Dataset info

Dataset ID	bigquery-public-data.london_bicycles
Created	May 25, 2017, 7:26:18 AM UTC-6
Default table expiration	Never
Last modified	Sep 20, 2022, 1:44:06 AM UTC-6
Data location	EU
Description	
Default collation	
Default rounding mode	ROUNDING_MODE_UNSPECIFIED
Case insensitive	false
Labels	
Tags	⚠️

3. Select the arrow to the left of the `london_bicycles` database name. This expands the dataset to reveal two table names: `cycle_hire` and `cycle_stations`. Select the `cycle_hire` table name within the **Explorer** pane.



Screenshot of BigQuery. The titles `cycle_hire` and `cycle_stations` are shown beneath `london_bicycles` in the explorer pane. `Cycle_hire` is selected.

This will pull the `cycle_hire` schema into the console. Take a moment to explore the field names and the associated information.

4. Now, select the **PREVIEW** tab to find a sample of the data that you'll be working with.

cycle_hire	QUERY	SHARE	COPY	SNAPSHOT	DELETE	EXPORT	REFRESH
SCHEMA	DETAILS	PREVIEW	LINEAGE	DATA PROFILE	DATA QUALITY		
Row	rental_id	duration	duration_ms	bike_id	bike_model	end_date	end_station_id
1	57870195	3840	3840000	4229		2016-08-31 20:49:00 UTC	null
2	57852555	3840	3840000	242		2016-08-31 15:19:00 UTC	null
3	57872531	3840	3840000	728		2016-08-31 22:12:00 UTC	null
4	57995603	2820	2820000	4375		2016-09-04 17:10:00 UTC	null

Once you have finished previewing the data, write a query!

Step 4: Review basic parts of a query

So far, you've learned three basic parts of a query: **SELECT**, **FROM**, and **WHERE**. As a refresher:

- **SELECT** is the section of a query that indicates what data you want SQL to return to you.
- **FROM** is the section of a query that indicates which table the desired data comes from. You must provide a full path to the table. The path includes the project name, database name, and table name, each separated by a period.

- **WHERE** is the section of a query that indicates any filters you'd like to apply to your table.

Step 5: Write a basic query

Now, construct a simple command using the basic parts of a query you have already learned! For example, you can select a specific column from the `cycle_hire` table, such as the `end_station_name` column.

1. Select the **Blue +** button or **QUERY - In new tab** to start a new query.
2. Start your query with a **SELECT** clause, and indicate which column you want to select from the table; in this case, you'll input `end_station_name`.
3. After you have indicated which column you are selecting, write your **FROM** clause. Specify the table you are querying from by inputting the following location:
``bigquery-public-data.london_bicycles.cycle_hire`;`

The completed query should appear like this:

1

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SELECT

end_station_name

FROM

`bigquery-public-data.london\_bicycles.cycle\_hire`;

4. Run your completed query by selecting the blue **RUN** button.

This query may take a few seconds to execute. Once it has finished, you will find the list of station names you requested under the **Query Results** console pane.

Step 6: Write a query to answer a question

After running the first basic query, try answering a specific question about the data. For example, how many bike trips lasted for 20 minutes or longer?

1. Select the **Blue +** button or **QUERY - In new tab** to start a new query. Start with your **SELECT** statement again. This time, include the two columns **duration** and **start_station_name** in the query. The data in these columns will tell where the trip started and the duration of the trip. Be sure to separate each column name with a **comma**.

2. Next, add your **FROM** statement. You will be using the same table as the previous query: **FROM `bigquery-public-data.london\_bicycles.cycle\_hire`;**

Note: The backticks around the table in this line of code are optional.

3. Finally, add a **WHERE** statement to specify that you want to filter for only bike rides 20 minutes or longer. If you check the preview of this data, you might notice that the

`duration` is recorded in seconds, so you'll specify 1200 seconds in your query.

Write that as `WHERE duration >= 1200;`

Your completed query should be written like this:

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```
SELECT
```

```
    duration,
```

```
    start_station_name
```

```
FROM
```

```
    `bigquery-public-data.london_bicycles.cycle_hire`
```

```
WHERE
```

```
    duration >= 1200;
```

4. Run your completed query by clicking the **RUN** button.

```
Untitled
[icon] RUN [icon] SAVE [icon] SHARE [icon] SCHEDULE [icon] MORE

1 SELECT
2   duration,
3   start_station_name
4 FROM `bigquery-public-data.london_bicycles.cycle_hire`
5 WHERE
6   duration >= 1200;
```

This query may take a few seconds to execute. Once it has finished, you will find a list of rides from this table that fit your criteria. There are millions of rows with bike trips that are 20 minutes or longer!

Optional Step 7: Up for a challenge?

If you're comfortable using queries to answer questions, try creating and running queries to complete the tasks below:

- What is the name of the station whose `start_station_id` is 111?
- Return all the `rental_ids`, station IDs, and station names that `bike_id` 1710 started from.
- What is the `bike_model` of `bike_id` 58782?

Step 8: Check your work

Use the solutions doc to check your work: [Intro to BigQuery Solutions](#)

Or download the file directly here:

[Intro to BigQuery solutions](#)

[DOCX File](#)

1.

Question 1

Reflection

Run another query on your table:

SELECT

end_station_name

FROM

`bigquery-public-data.london\_bicycles.cycle\_hire`

WHERE

rental_id = 57635395;

At what station did the bike trip with `rental_id` 57635395 end?

Southwark Street, Bankside

Tower Gardens, Tower

East Village, Queen Elizabeth Olympic Park

Answer: Notting Hill Gate Station, Notting Hill

Correct

The address listed under the end_station_name column for Row 1 of your results table was East Village, Queen Elizabeth Olympic Park. To find this, you successfully ran a query. Going forward, you will continue using **SELECT**, **FROM**, and **WHERE** statements in your queries to interact with databases using SQL. This will help you build more complicated SQL queries when you are analyzing data in the future.

1 / 1 point

2.

Question 2

In this activity, you had an opportunity to get more familiar with BigQuery and writing SQL queries. In the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- How do you think you can use public datasets in BigQuery to help develop your data analysis skills?
- How do you think understanding basic query syntax will help you write more complex queries in the future?

Yes, I'm very pleased about learning this step of BigQuery. I'm very happy to learn BigQuery, and my heart becomes full of joy after learning the query. Yes, obviously now I'm able to use public datasets in BigQuery to help develop my data analysis skills.

Yes, the basic query syntax is the base of the pillar to write basic BigQuery commands. Now, I'm more confident about BigQuery, and I've generated the basic idea of BigQuery. And of ofcourse I think understanding basic query syntax will help me write more complex queries in the future.

Feedback

Congratulations on completing this hands-on activity! You explored BigQuery, uploaded public data to your console, and constructed some queries. An effective response would note that BigQuery public datasets help you practice writing SQL. In upcoming activities, you will continue exploring working with databases and writing queries with SQL—an essential tool in every data analyst's toolkit.